



NATIONAL CENTER FOR HEALTH STATISTICS

Technical Notes

Round 5: Data collected July 2024



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Introduction

The National Center for Health Statistics (NCHS) Rapid Surveys System (RSS) is a platform that utilizes commercially available probability-based online panels to provide time-sensitive data about emerging and priority health concerns. RSS began fielding in 2023, and has a different questionnaire administered each round of data collection.

To provide timely access to selected point estimates based on RSS content, a dashboard may be released following each round of data collection. Percentages are shown by selected population subgroups such as age group, gender, race and Hispanic origin, education, household income as a percentage of the federal poverty level, region, and urbanicity.

Methods

Data source

The target population of RSS Round 5 (RSS-5) is children under the age of 18; panel members were only eligible to participate if they were the adult parent or legal guardian of a child under 18 who lived in their household. Data were collected in July 2024 from two commercial panel providers using the same questionnaire (www.cdc.gov/nchs/data/rss/round5/questionnaire.pdf). Data were collected for about 8,101 children using two panels – Amerispeak (conducted by NORC at the University of Chicago) and KnowledgePanel (conducted by Ipsos). The combined completion rate for both panels was 37.3 (1).

Both panel providers collect profile information from their panelists on a regular basis, including several sociodemographic and geographic characteristics. As these data were already available for RSS respondents as part of their panel profile data, questions about these characteristics were not re-asked on the RSS questionnaire. These measures were harmonized into common categories, but the information was collected separately from RSS, at different times and using different questions in each panel.

Details on data collection, sampling methods, response rates, weighting methodologies, and other data processing components can be found in the Survey Description (www.cdc.gov/nchs/data/rss/round5/survey-description.pdf) and the Quality Profile (<https://www.cdc.gov/nchs/data/rss/round5/quality-profile.pdf>).

Estimation procedures

RSS is a sample survey conducted by a commercial survey panel. That is, only a sample (subset) of the civilian noninstitutionalized population is invited to join the panel and a sample of those panelists are selected to participate in the survey. Additionally, not everyone invited to join the panel or selected to participate in the survey agrees to participate, which can affect the representativeness of the sample. To account for these factors, sampling weights are created. These sampling weights are used to produce representative national estimates. The data must be weighted to obtain population estimates for survey outcomes in the population represented by the RSS. The value of the weight for a given respondent can be interpreted as the number of persons in the RSS target population represented by that respondent.

The RSS-5 questionnaire included several variables which were used to calibrate survey weights to NHIS population totals. These final calibrated weights were used to generate the estimates in the released dashboard. All estimates shown meet the [NCHS Data Presentation Standards for Proportions](#).

Cases with missing data are excluded from the analysis, unless otherwise mentioned. Data would be considered missing for a variable if, for example, the respondent refused or didn't know how to respond, or if they skipped the question on the web. These are henceforth referred to as nonresponse. Overall, item nonresponse rates were low, averaging <2% per item in the combined file. Several of the sociodemographic characteristics used in the dashboard have very low or no missing values, as they were imputed when missing for weighting purposes. None of the variables related to measures in the RSS-5 dashboard had an item nonresponse higher than 5% in the combined dataset. Data users using variables on the combined file with higher item nonresponse are encouraged to consider the impact of nonresponse, if applicable to their analysis/research. Please see the Quality Profile (<https://www.cdc.gov/nchs/data/rss/round5/quality-profile.pdf>) for more information.

Data limitations

While faster to produce compared to traditional household surveys, web-based panel surveys are subject to issues regarding accuracy and usability. Web-based panel surveys often have lower response rates than large-scale national surveys and may underrepresent certain subpopulations. This can lead to biased estimates when attempting to make inferences to the national population. RSS's smaller sample sizes and greater potential for coverage and nonresponse bias may result in lower precision, especially for subgroups. Panel survey nonresponse occurs at many stages, including panel recruitment, panel retention, and at the individual survey level. RSS aims to compensate for nonresponse through calibration and weighting of the RSS to gold standard NCHS surveys. However, the effectiveness of these weighting adjustments for

nonresponse may vary across survey estimates and will depend on the availability of appropriate gold standard survey data. RSS also includes a benchmarking component, which allows comparison of estimates to those from the same questions on other surveys, to facilitate bias assessments for a wide array of health-related estimates. These bias assessments provide context on the effectiveness of the weighting adjustments and quality of estimates generated from RSS. For an evaluation of the quality of RSS-5 data, including the calibration of weights and benchmark analysis, please see the Quality Profile (<https://www.cdc.gov/nchs/data/rss/round5/quality-profile.pdf>).

Another limitation of RSS is that some of the sociodemographic and geographic variables are drawn from panel profile variables, which are collected separately from RSS, and are collected at different times than the RSS health topic content and using different questions in each panel. Although they are updated regularly, it is not known whether any of these characteristics had changed between the last time the panel collected the information and the respondent completed the RSS-5 questionnaire.

The Rapid Surveys System is particularly well suited for time sensitive data needs, measuring public health attitudes, developmental work to improve concept measurement, and methodological studies, but is intended to complement and not replace the current household survey systems at NCHS, including the National Health Interview Survey.

Variance estimation and statistical reliability

All estimates shown meet the NCHS standards of reliability as specified in *National Center for Health Statistics Data Presentation Standards for Proportions* (2). Unreliable estimates are indicated with an asterisk (*) and are not shown. Reliable estimates with an unreliable complement are shown but are indicated with two asterisks (**). Complements are calculated as 100 minus the percentage. The standards are applied directly for percentages. Two-sided 95% confidence intervals are calculated using the Clopper-Pearson method adapted for complex surveys by Korn and Graubard (2). Standard errors used in this calculation were obtained using SUDAAN software, which takes into account the complex sampling design of RSS. The Taylor series linearization method was used for variance estimation.

Definitions of selected terms

Sociodemographic and geographic characteristics from panel profile data

The following sociodemographic and geographic characteristics used as covariates in these dashboards and tables were collected as part of the panel profile information (not RSS questionnaire) and harmonized between the two panels.

Household income as a percentage of the federal poverty level – Categories presented are Less than 100% FPL, 100% to less than 200% FPL, and 200% FPL and greater. FPL is federal poverty level.

Region – In the geographic classification of the U.S. population, states are grouped into four regions used by the U.S. Census Bureau:

Northeast: Connecticut, Maine, Massachusetts, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, and Vermont

Midwest: Illinois, Indiana, Iowa, Kansas, Michigan, Minnesota, Missouri, Nebraska, North Dakota, Ohio, South Dakota, and Wisconsin

South: Alabama, Arkansas, Delaware, District of Columbia, Kentucky, Florida, Georgia, Louisiana, Maryland, Mississippi, North Carolina, Oklahoma, South Carolina, Tennessee, Texas, Virginia, and West Virginia

West: Alaska, Arizona, California, Colorado, Hawaii, Idaho, Montana, Nevada, New Mexico, Oregon, Utah, Washington, and Wyoming.

Urbanicity – Based on the 2013 NCHS Urban-Rural Classification Scheme for Counties (3) which groups U.S. counties and county-equivalent entities into six categories: large central metropolitan, large fringe metropolitan, medium metropolitan, small metropolitan, micropolitan, and non-core. For the RSS dashboards, medium and small metropolitan are combined into a single group, and micropolitan and non-core are combined into a single group (nonmetropolitan).

Sociodemographic characteristics collected on the questionnaire

The following sociodemographic characteristics used as covariates in this dashboard were collected on the RSS-5 questionnaire.

Age – Age of the child is recorded in single years (with the option to record age in months for children under 2 years old) and grouped into categories for the dashboard.

Gender – Children are classified as Male or Female.

Race and ethnicity – Per guidance from OMB ([Federal Register :: Revisions to OMB's Statistical Policy Directive No. 15: Standards for Maintaining, Collecting, and Presenting Federal Data on Race and Ethnicity](#)), starting in RSS-5, race and ethnicity were collected using different questions than previous rounds. Two different categorizations of race and ethnicity are presented in the dashboard.

Race and ethnicity – combined Multiracial and/or Multiethnic approach. This categorization presents the following mutually exclusive categories: American Indian or Alaska Native alone; Asian alone; Black or African American alone; Hispanic or Latino alone; Middle Eastern or North African alone; Native Hawaiian or Pacific Islander alone; White alone; Multiracial and/or Multiethnic.

Race and ethnicity – alone or in combination approach. This categorization presents the following potentially overlapping categories: American Indian or Alaska Native alone or in combination; Asian alone or in combination; Black or African American alone or in combination; Hispanic or Latino alone or in combination; Middle Eastern or North African alone or in combination; Native Hawaiian or Pacific Islander alone or in combination; White alone or in combination.

Select outcomes collected on questionnaire

Estimates for most measures are generated from a single question on the questionnaire. For exact wording of questions, please see the questionnaire (www.cdc.gov/nchs/data/rss/round5/questionnaire.pdf). Below are additional details on some of the outcomes shown in the dashboard which may be more complicated to derive.

Attitudes on Childhood Vaccines –

A series of questions asked which of the following had communicated with the parent about getting vaccines for their child: a doctor or other health care provider, school or daycare, some other source. Respondents could have selected more than one source.

A question asking how difficult the parent thought it was to get their child vaccinated had the following answer categories: Not at all difficult, Somewhat difficult, Very difficult, and I have not tried to get [child] vaccinated. The measure in the dashboard includes those who responded they had not tried to get their child vaccinated in the denominator.

Parents who responded that it was somewhat or very difficult to get their child vaccinated to the previous question were then asked about how much they agree or disagree with possible reasons why this vaccination was difficult. Possible reasons

included lack of reliable transportation, because of the cost, and it is hard to find the time to take the child to get vaccinated. Respondents could have selected more than one source. Each of these questions had the following answer categories: Strongly agree, Somewhat agree, Somewhat disagree, and Strongly disagree.

Positive Childhood Experiences –

A question asked about how much the parent agrees that teachers at their child's current school care about their child. This question was asked about children ages 4 and older who attended public, private, or charter school in-person, attended a virtual school full time, or attended a college or university, and excluded those who had not yet started school, as well as those who were homeschooled full-time. This question had the following answer categories: Strongly agree, Somewhat agree, Somewhat disagree, and Strongly disagree.

A series of questions asked which activities the parent has done with their child in the last seven days. These activities included whether they read books or told stories together, cooked or enjoyed meals together, educational activities, spent time outdoors including walks and sports, watched TV or other media together, and for parents of children aged 3 and older, played video games together, or played board or card games together. Children ages 0-2 whose parents were not asked the last two questions in this series were counted as not having engaged in those activities. Otherwise, this measure excluded children with at least one of these questions which was skipped or had responses of refused or don't know.

A question asking about how difficult it is for children aged 6 and older to make friends had the following answer categories: No difficulty, Some difficulty, A lot of difficulty, or Cannot do at all. 0.8% of children aged 6 and older were reported to not be able to make friends at all, and were not asked follow-up questions such as how often children play or hang out with friends in person in an average week.

Further information

Data users can obtain the latest information about RSS by periodically checking the website (<https://www.cdc.gov/nchs/rss/rapid-surveys-system.html>). This website will feature downloadable public-use data and documentation for RSS, as well as important information about any modifications or updates to the data or documentation.

References

1. National Center for Health Statistics. Rapid Surveys System (RSS): Round 5 survey description. 2024.
Available from: www.cdc.gov/nchs/data/rss/round5/survey-description.pdf.

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2. Parker JD, Talih M, Malec DJ, Beresovsky V, Carroll M, Gonzalez JF Jr, et al. National Center for Health Statistics data presentation standards for proportions. National Center for Health Statistics. Vital Health Stat 2(175). 2017.
 3. National Center for Health Statistics. NCHS urban–rural classification scheme for counties. Available from:
[https://www.cdc.gov/nchs/data_access/urban_rural.htm#2013 Urban-Rural Classification Scheme for Counties](https://www.cdc.gov/nchs/data_access/urban_rural.htm#2013_Urban-Rural_Classification_Scheme_for_Counties).

Suggested citation

Recommended citations for specific tables and charts are included in the notes at the end of each page. The citation for the Technical Notes is as follows, although it should also include the date accessed as it may be edited periodically when new tables are added.

Technical Notes. NCHS Rapid Surveys System. Round 5. December 2024. National Center for Health Statistics. Available from: www.cdc.gov/nchs/data/rss/round5/technical-notes.pdf.